

AIRSTREET · PROBLEM STATEMENT

AirStreet problem statement

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 update; constraints and open questions reflect AST-DDR-001 and AST-CAL-001
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (problem, users, context, constraints, prior work)
0.1	2026-09-25	Amish Chadha	Initial scaffold

AirStreet problem statement

Regulatory air monitors are few and far apart, missing the streets where people breathe traffic pollution.

■ The problem

Traffic pollution varies sharply over short distances, but official monitoring is designed to describe whole cities. When Google Street View cars carrying reference-grade instruments drove every street in a 30 km² area of Oakland, California, nitrogen dioxide (NO₂), nitric oxide and black carbon varied by up to 5 to 8 times within single city blocks (Apte et al., 2017). A single fixed station cannot show that pattern, and many cities have no station at all. Only 12 % of India's 4,041 census cities and towns have any air quality monitoring (CSE, 2023), and in 2020 OpenAQ found no public national government monitoring program in 13 populous countries, including Pakistan, Nigeria, Ethiopia, Kenya and Uganda (OpenAQ, 2020).

The health stakes are large. Ambient air pollution caused an estimated 4.2 million premature deaths worldwide in 2019 (WHO). NO₂ comes mainly from burning fuel in cars, trucks, buses, power plants and off-road equipment (US EPA), so it is highest on busy streets, and more than 45 million people in the United States live within about 90 m (300 ft) of a major transportation facility (US EPA).

Low-cost sensors could fill the gap between stations, but raw readings are not trusted. Uncorrected PurpleAir particle sensors overestimated PM_{2.5} by about 40 % across the United States until a humidity-aware correction cut the error from 8 to 3 µg/m³ (Barkjohn et al., 2021). Electrochemical NO₂ sensors respond to temperature, humidity and other gases, and laboratory calibration alone does not correct this; in a Pittsburgh study only field-calibrated models reached a mean absolute error of about 3.5 ppb for NO₂ (Zimmerman et al., 2018).

AirStreet addresses one specific gap: an open, pole-mounted node that measures PM_{2.5} and NO₂ at street level, with a calibration method stated up front, so a city, school or community group can build a neighborhood map that others can check.

■ Users and context

User	Need	Context
Residents and community groups	See pollution on their own streets, near schools and clinics, and track whether it changes	Street poles on residential and arterial roads
Municipal environment and transport teams	Evaluate interventions such as low-traffic streets, school streets, bus corridors and freight routes	10 to 100 nodes per district, alongside at least one reference station
Schools and universities	Teaching and research on exposure, with raw data and a documented method	Campus and nearby streets
Street furniture owners (utilities, city lighting)	A small, light device that does not interfere with the pole or its wiring	Street light and signal poles
Researchers and data platforms	Raw signals and calibration versions, so data can be reprocessed	Open data portals

■ Operating environment

- Mounted on a street light or signal pole, inlets 3.0 m above the sidewalk (AST-DDR-001 D7, decided by Amish on 2026-09-25), on the street side and within a few meters of the kerb.
- Outdoor, in sun, rain, dust and traffic spray; temperate to hot humid climates (about -10 to 50 °C, 10 to 100 % RH, estimate to be confirmed with partners). FieldNode is rated to 45 °C ambient, so sites hotter than that need FieldNode's sun shield (AST-DDR-002); the SPS30 particle sensor is recommended for 20 to 80 % RH, so the most humid sites remain at risk (AST-CAL-001).
- No mains power: runs on the FieldNode solar and battery core.
- Data over LoRaWAN: every 5 min through a private gateway (the lab's TwinKit gateway by default), or every 15 min on The Things Network, whose fair-use policy a 5 min interval would exceed (AST-DDR-002).

■ Constraints

- Garage-buildable prototype. The \$285 `budget_usd` covers the AirStreet sensor head; the FieldNode core is costed in FieldNode (AST-DDR-001 D1 and AST-DDR-002). The sensor head costs \$285.00 at TRL 3 prices, within the budget with no margin (see [03-requirements.md](#)).
- Built on FieldNode, the lab's shared outdoor core, with its power budget and sensor port pinout.
- Temperature, humidity and particle checks on CalRig. CalRig does not cover NO₂, so NO₂ must be calibrated by field collocation with a reference station.
- Levels only: no images, audio or personal data collected or sent.
- Mounting only with the pole owner's permission, without drilling the pole or opening its electrical access.

■ Out of scope

- Regulatory compliance monitoring. AirStreet data are indicative and do not replace reference stations.
- Other pollutants (ozone, carbon monoxide, black carbon, ultrafine particles) in this version.
- Personal exposure devices and indoor use.
- Health advice or medical claims.

■ Prior work

- Street-scale mapping with reference instruments on vehicles showed block-level variation ([Apte et al., 2017](#)).
- Community and city networks of low-cost sensors already operate, for example AirQo, which reports more than 400 PM_{2.5} sensors across 14 African countries ([AirQo](#)), and Breathe London, a sensing network developed with the Mayor of London ([Breathe London](#)).
- The US EPA published performance testing protocols for NO₂, CO and SO₂ sensors in February 2024, including field evaluation alongside regulatory monitors ([US EPA](#)).
- Humidity correction for optical particle sensors ([Barkjohn et al., 2021](#)) and field calibration of electrochemical gas sensors ([Zimmerman et al., 2018](#)) are established methods that AirStreet adopts rather than reinvents.

What is missing is an open, documented node design that pairs these sensors with a stated, repeatable calibration route and publishes raw signals.

■ Open questions

- Which city and partner first, and which reference station can host a collocation? Proposed, awaiting Amish (AST-DDR-001 O1).
- Inlet height: 3.0 m is decided (AST-DDR-001 D7); partners may still ask for a lower inlet, closer to the breathing zone, which would reopen that decision.
- Is NO₂ accuracy of about 4 to 5 ppb hourly, which cannot resolve the WHO annual guideline but can rank streets, useful enough for the first partner's questions? To be tested with users. Resolving the WHO guideline is no longer a requirement; it is kept as a research question (AST-DDR-002).

■ User research and co-design

This design is for communities the author is not part of, so requirements come from the people who will use it.

- [] Identify a local partner organization (Helpful Engineering network, NGO or university)
- [] Run co-design sessions with intended users; record who, where and what was learned
- [] Validate load, distance, terrain and cost assumptions in the field
- [] Revise requirements (REQ) from findings before freezing the design